

Photo-crosslinked copolymers of 2-hydroxyethyl methacrylate, poly(ethylene glycol) tetra-acrylate and ethylene dimethacrylate for improving biocompatibility of biosensors

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A copolymer containing 88% 2-hydroxyethyl methacrylate (HEMA), 9% poly(ethylene glycol) (MW 18.5 kDa) tetra-acrylate and 3% ethylene dimethacrylate was prepared and evaluated for use as a biocompatible interface between glucose biosensors and tissue in the rat. The glucose sensor utilizes glucose oxidase that is electrically 'wired' to a gold current collector by a reduction–oxidation polymer. Coatings of the copolymer were crosslinked *in situ* on the sensors using long wavelength ultraviolet light and 2,2-dimethoxy-2-phenyl-acetophenone as the initiator. The effect these films had on the current response to glucose was measured. Over a glucose concentration range of 0–30 mM, the average percentage decrease in response was $45 \pm 28\%$ (mean $\pm 95\%$ confidence interval) at 37°C for films that were about 0.1 mm thick, an acceptable value. Copolymer-treated and control electrodes were implanted in the intrascapular subcutaneous tissue of male Sprague-Dawley rats for three days. The explanted samples were evaluated using scanning electron microscopy. The control electrodes were highly encapsulated with fibrous material, while the copolymer-treated electrodes induced much less encapsulation. The results show this copolymer to be a candidate as a biocompatible coating for electrically wired oxidoreductase-based subcutaneous biosensors.

Keywords: Biocompatibility, biosensors, subcutaneous, poly(ethylene glycol), glucose oxidase

Received 19 January 1994; accepted 16 June 1994

Although *in vivo* sensing of glucose has received considerable attention, little work has been done on the design of materials specifically for the sensor–tissue interface. Some stringent requirements, including absence of protein fouling, cellular and fibrous overgrowth, thrombus formation, inflammation and tissue necrosis, must be met. In addition, it is desirable in some sensors to have an interface that is highly permeable to glucose. The sensors tested for this work were based on electrically wired glucose oxidase. These sensors had no diffusional mediators that might leach into the tissue and did not require O₂ for their operation¹. Instead, reduction–oxidation (redox) polymers were used to wire the enzyme electrically to the current collector².

In most biosensors currently under development, the materials used are frequently chosen for their electri-

cal, mechanical or electrochemical properties. One result is that these materials may not elicit the desired biological response when implanted in the host and untoward host responses may compromise the integrity of the measurement. This work was aimed at improving the biocompatibility of the biosensor. Materials for use in coating the biosensor must have the following characteristics:

1. The material must adhere well to the outer material of the sensor. This may be wire insulation, a layer of the sensing material or a layer of some other material.
2. The material must not be highly deleterious to the activity of the enzyme used in either its applied (e.g. a monomer, prepolymer or polymer solution) or final state.
3. The material must allow the transport of the analyte and the products of its transformation.

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4. The material must elicit the desired biological response in the host.

Other researchers have previously used a variety of membrane materials, including hydrophilic polyurethanes^{3,5}, polycarbonates^{6,7}, cellulose-based formulations⁸, silicone rubbers^{9,10}, fluorinated ionomers¹¹, and poly(vinyl alcohol)-heparin hydrogels¹². These materials have been applied by several means. Several investigators have utilized casting, directly on the sensor, from organic solvents. Bindra *et al.*³, Poitout *et al.*⁴ and Koudelka *et al.*⁵ reported solvent casting of polyurethanes from solutions of tetrahydrofuran and dimethylformamide. They reported good retention of enzyme activity after treatment with the polymer solution and acceptable performance *in vivo*. However, no biological analysis of explanted sensors was discussed. Turner *et al.*¹¹ reported solvent casting of a fluorinated ionomer (NafionTM) from a solution of water and lower aliphatic alcohols. They performed both long- and short-term implant experiments. Scanning electron microscopic and histological analysis of the implants revealed no signs of acute or chronic foreign body response. For the short-term implants (4 days), fibroblasts and other cell types including macrophages and monocytes were evident on the surface. Long-term implants were encapsulated in fibrous tissue that was found to be approximately 10–100 μm thick. It should be noted that they did not apply these films to any functioning biosensors.

Others have used reactively cured systems. Gough *et al.*⁹ and Armour *et al.*¹⁰ utilized systems of silicone rubber and silicone rubber/albumin, respectively, that were crosslinked *in situ* on the sensors. Both indicated that the sensors responded to glucose in dogs after treatment with the polymers. Armour reported that one electrode was encapsulated with a thin layer of tissue when retrieved. Brinkman *et al.*¹² reported on the use of photo-crosslinked poly(vinyl alcohol)-heparin hydrogels. The films were not applied to any sensors and no *in vivo* testing was reported. Finally, some researchers utilized preformed mechanical membranes on enzyme sensors. Keedy and Vadgama⁶ and Mascini *et al.*⁷ used commercially available polycarbonates that they silanized by exposure to the vapours of mixtures of dimethyldichlorosilane and 1,1,1-trichloroethane to cover their enzyme-based sensors. Both reported acceptable retention of enzyme function after covering with the films. Keedy and Vadgama did no *in vivo* testing. Mascini *et al.* implanted their sensors subcutaneously in humans. However, they did no biological analysis on the explanted sensors. Pickup *et al.*⁸ used a cellulose dialysis membrane that was stretched over their glucose sensors. Their sensors also retained good enzyme activity and the sensors responded to glucose when implanted subcutaneously in humans. They did not discuss biological analysis of the explanted sensors. Turner *et al.*¹¹ also tested preformed Nafion discs. These preformed membranes were not applied to any sensing devices and the results of their *in vivo* tests were already discussed (same as for the solvent cast fluorinated ionomer membranes).

The use of poly(ethylene glycol) (PEG) to reduce

protein adsorption and encapsulation by fibrous tissue has been thoroughly investigated^{13–17}. Because of its good biological performance in other systems, we decided to extend the use of this polymer to biosensors. Our approach was to crosslink *in situ* a comonomer that contains a significant amount of PEG on the surface of the sensor. This comonomer mixture was approximately 88% HEMA, 9% tetra-acrylated PEG (PEG-TA; MW 18 500 g mol⁻¹) and 3% ethylene dimethacrylate. The crosslinked copolymer was synthesized upon the sensor tip using ultraviolet light (360 nm) and 2,2-dimethoxy-2-phenyl-acetophenone to initiate the free radical reaction. After hydration, this material swelled to form a hydrogel, allowed free transport of glucose and the products of its oxidation and displayed excellent biological properties when implanted subcutaneously in Sprague-Dawley rats for three days, the intended duration of use for this sensor.

MATERIALS AND METHODS

Glucose sensor preparation

The glucose sensors used for this study, illustrated in Figure 1, were fabricated using polyimide-insulated 250 μm diameter gold wire (California Fine Wire, Grover City, CA, USA). First, 5–8 cm lengths of the wire were cut and electrical connection to an 8–10 cm length of 30 AWG wire (stainless steel) was made using silver epoxy (Epo-Tek H20E; Epoxy Technology, Billerica, MA, USA). The epoxy was applied to one end of the gold wire and to a stripped end of the 30 AWG wire. A heat-gun was used to cure the silver epoxy. A short length of 3/64-inch heat shrinkable tubing (Raychem, Menlo Park, CA, USA) was placed over the junction of the two wires and shrunk by heating with air from a heat-gun. Approximately 125 μm of the gold was then electrochemically etched away to leave a cavity that was subsequently filled with a multilayer polymer-enzyme system, including the sensing layer that transduces the glucose concentration to current and an interference eliminating layer¹⁸.

The contents of these layers were as follows:

Layer 1. The innermost sensing layer was a heavily crosslinked macromolecular network formed of 0.20 μg of genetically engineered glucose oxidase (Chiron, Emeryville, CA, USA), 0.31 μg of a complex between polyvinyl imidazole and osmium (2,2-bis(bipyridine chloride) cation (the redox polymer) and 0.07 μg of poly(ethylene glycol) diglycidyl ether. The redox polymer was derivatized with ethyl amine pendant

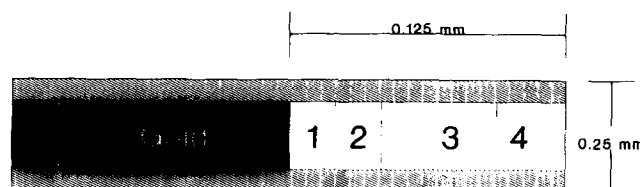


Figure 1 Scheme of the four-layered glucose sensor. Cross-hatch indicates polyimide insulation. 1, Sensing layer; 2, insulating layer; 3, interference-eliminating layer; and 4, biocompatible layer.

groups. The diglycidyl ether covalently bonds the redox polymer to the enzyme through the primary amine groups on each.

Layer 2. The insulating layer was a crosslinked matrix of 0.03 μg of polyallylamine and 0.32 μg of a polyaziridine. The purpose of this layer is to eliminate unwanted 'wiring' of the enzymes lactate oxidase and horseradish peroxidase that are in the interference eliminating layer.

Layer 3. The interference eliminating layer was a crosslinked polymer made by oxidizing the oligosaccharide of horseradish peroxidase (0.39 μg) to its aldehyde and crosslinking it with the lysine amine residues of lactate oxidase (2.96 μg)¹⁹.

Layer 4. This tissue-contacting layer consists of approximately 0.19 μg of the photopolymerizable monomer and prepolymer mixture, cured *in situ* to form a hydrogel membrane.

Polymer film preparation

A tetrahydroxy PEG, formed by linking two PEG 9000 diol chains via a bisphenol A bisepoxide (Polysciences, PEG 18 500 g mol^{-1}), was acrylated. Acryloyl chloride (Aldrich, Milwaukee, WI, USA) was vacuum distilled. Triethyl amine (Mallinkrodt, Paris, KY, USA) was dried with molecular sieves. A 500 ml reaction flask and a reflux condenser were assembled in a heating mantle over a stir plate. The glassware was flame dried under vacuum and purged with dry argon. PEG (20 g) was added to the reaction vessel along with 200 ml of benzene, to make a 10% solution (w/v) of the polymer. A slight amount of heating while the mixture was stirring was required to dissolve the polymer (10–15 min at 40–50°C). The apparatus was removed from the heating mantle and allowed to cool to room temperature. The reaction flask was then placed in an ice bath over the stirring plate. Once cooled, 0.7 ml dry triethyl amine and 1.13 ml of acryloyl chloride were added (4 $\times$ molar excess acryloyl chloride, with equimolar triethyl amine). The reaction vessel was removed from the ice bath and returned to the heating mantle. The mixture was heated and allowed to reflux at approximately 79°C for 2 h. The reaction flask was stirred overnight under argon. The reaction is schematically represented in Figure 2. The amine salts were subsequently removed by filtration with a Buchner funnel. The tetra-acrylated PEG (PEG-TA) was precipitated in 2 l of reagent grade hexane. The product was filtered and dried overnight in a vacuum oven at 40°C.

HEMA (Sigma, St. Louis, MO, USA) and the ethylene dimethacrylate (Sigma) were vacuum distilled before use. The initiator used for the crosslinking step was 2,2-dimethoxy-2-phenyl-acetophenone (Aldrich). A solution of 300 mg ml^{-1} of the initiator in 1-vinyl-2-pyrrolidinone (Aldrich) was made and added to the monomer mixtures. A prepolymer solution was made by adding 200 mg of the PEG-TA and 60 μl of ethylene dimethacrylate to 2 ml of the HEMA. Approximately 30 μl of the initiator mixture was added to the monomer-prepolymer (final initiator concentration 3.93 mg ml^{-1}).

The copolymer film was crosslinked *in situ* inside

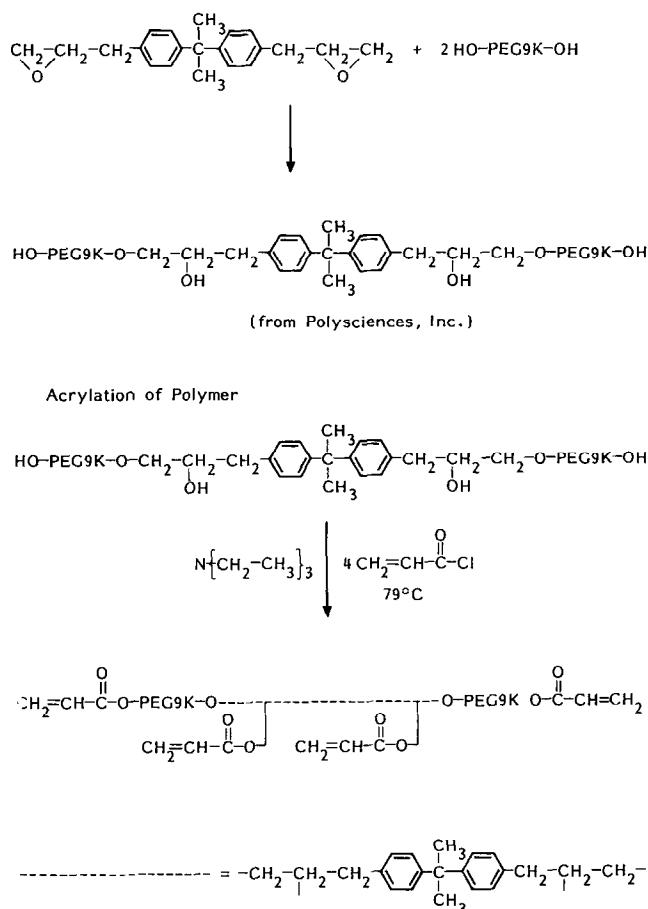


Figure 2 Chemistry of acrylation of poly(ethylene glycol) tetraacrylate. Product from Polysciences and synthesis scheme are shown.

the cavity of the electrode. Approximately 12 nl of the prepolymer solution was added to the cavity using a micropipette, pulled from a capillary tube, connected to a microsyringe with polyethylene tubing and shrink wrap, and a micromanipulator. The process was visualized using a stereo microscope (Bausch & Lomb, $\times 30$ WF, eyepieces with $\times 4$ zoom capability). The sensor was then exposed to 360 nm UV light for up to 2 min. The light source was a mercury vapour arc lamp with a UV filter (peak intensity of 360 nm and power output of about 200 mW cm^{-2} ; Blak-Ray; UVP, San Gabriel, CA, USA). The electrodes so modified were soaked in phosphate-buffered saline (PBS; 32 g NaCl, 0.8 g KCl, 8.68 g Na₂HPO₄·7H₂O and 0.8 g KH₂PO₄ in 4 l of deionized water, pH 7.4) for approximately 1 h before being calibrated. The four-layer sensor is schematically represented in Figure 1.

Attenuated total reflectance Fourier transform infrared spectroscopy (ATR-FTIR) was used to study the degree of double bond conversion in the film as a function of duration of exposure to UV light. A thin film of the monomer-prepolymer-initiator solution was spread onto the top surface of the ATR crystal. After the necessary background scans were recorded, the film was exposed to the UV light source. Scans were recorded every 6 s for up to 10 min. We examined the 1630–1650 cm^{-1} region. This region contained the

strongest absorbing carbon-carbon double bond peaks. Subtraction of the background scans from each subsequent scan yielded difference spectra as a function of duration of exposure, providing an approximate measure of double bond conversion. The percentage double bond conversion was calculated by comparing the intensity values at $t = 0, 2$ and 10 min. No further change in peak intensity was observed after 10 min of UV exposure.

The copolymer film was further characterized by evaluating the degree of swelling in phosphate-buffered saline. Films of the copolymer were cast onto a glass sheet and hydrated. The film was then cut into pieces approximately $5\text{ mm} \times 5\text{ mm} \times 2\text{ mm}$. These samples were then dried in a vacuum oven overnight (60°C , 25 mm Hg). Subsequently, the samples were soaked in phosphate-buffered saline for 3 h . The samples were removed from the solution, dried of excess fluid with a tissue and weighed. The ratio of water weight to total weight was calculated.

Glucose calibration testing

Sensors were tested for their amperometric response to glucose after initial preparation and after any modifications were made. This was accomplished by altering the concentration of glucose in the PBS test solution from 0 to 20 or 30 mM , in 2 or 5 mM increments, by adding aliquots of 2 M glucose. The test cell used was a jacketed 10 ml cell that was heated to 37°C . During the calibration testing, the sensor was poised at a potential of 300 mV using a bipotentiostat.

Response to UV light exposure

Since the enzyme glucose oxidase has absorption peaks in the spectrum of the UV lamp (absorption peaks at $280, 350$ and 450 nm), the effect of exposure to UV light upon the amperometric response of the sensor to glucose was determined. This was accomplished by calibrating the sensor, as described previously, then exposing it to the UV light source for various exposure times and recalibrating the electrode. The effect of UV light exposure was then calculated as a percentage decrease in sensor output at each glucose concentration.

Response to prepolymer-monomer mixture exposure

Glucose oxidase, like many enzymes, is sensitive to organic environments. Since the monomer-prepolymer mixture contains organic constituents, the fraction of the total decrease in sensor performance due to exposure to the prepolymer mixture was determined. The electrode was calibrated and the prepolymer was then injected into the cavity at the tip of the sensor until it was full. The sensor was then held at ambient conditions for approximately 2 min and recalibrated. The effect of this exposure was calculated as a percentage decrease in sensor output at each analyte concentration. The change in sensitivity of the sensor was also calculated from the slope of the current versus glucose concentration curve. Only the $0\text{--}15\text{ mM}$ linear range was used for this calculation.

Implant preparation and procedure

Control electrodes which had only layers 1 through 3 and treated electrodes which contained all four layers were cut to lengths of about 1 cm . Both types of implants were sterilized by exposure to ethylene oxide gas overnight in a room temperature ethylene oxide sterilizer. The electrodes were removed from the sterilization pouches and soaked in sterile PBS prior to implantation.

The sensors were implanted subcutaneously in adult Sprague-Dawley rats. The rats were anaesthetized with halothane and additional halothane was administered, as needed, during the course of the short procedure. The intrascapular area of the rat was shaven and cleaned with an alcohol swab. A small incision was made in the skin using sterile surgical scissors. The cut was made deep enough to penetrate all layers of the skin. A stainless steel probe was used to open a cavity in the subcutaneous tissue. The sterile sensor was inserted into this cavity with sterile forceps. The wound was then closed by suturing with $4\text{--}0$ silk.

Processing of explanted samples

After the rats were allowed to return to their normal environment for three days, they were killed by carbon dioxide asphyxiation. The implants were immediately retrieved and placed in PBS.

The samples were fixed using buffered formaldehyde and step-wise dehydrated in ethanol ($60\text{--}100\%$). The samples were then soaked in a $50:50$ mixture of 100% ethanol and Peldri IITM (Ted Pella, Redding, CA, USA) and subsequently in two baths of 100% Peldri II. The samples were then refrigerated and the Peldri II was removed by vacuum sublimation. Samples were stored in a desiccator until they were sputtered with gold. The resulting specimens were examined using a Phillips 515 scanning electron microscope.

RESULTS AND DISCUSSION

The prepolymer formed a three-dimensional network in less than 1 min of UV illumination. To examine the effect of exposure to UV light for durations of this order, sensors were exposed to the 360 nm light source for $0.5, 1, 2, 3$ and 4 min . Each sensor was calibrated before and after each exposure. Results are shown in Figure 3. In electrode fabrication, a UV exposure time of up to 2 min was used to ensure complete crosslinking. For 2 min UV exposure, the overall decrease in output was estimated to be $20 \pm 5\%$ (mean $\pm 95\%$ confidence interval).

Calibrated sensors were also exposed to the monomer-prepolymer mixture, without crosslinking, for approximately 2 min . The electrodes were then recalibrated and the percentage decrease in sensor response to glucose was calculated. The results and statistical analysis for several electrodes are shown in Table 1. The current responses of one typical electrode are illustrated in Figure 4. The average percentage decrease in amperometric response to glucose over the concentration range $0\text{--}30\text{ mM}$ was $15 \pm 12\%$. The corresponding change in sensitivity was 1.32 ± 0.76 to $1.07 \pm 0.53\text{ nA mM}^{-1}$.

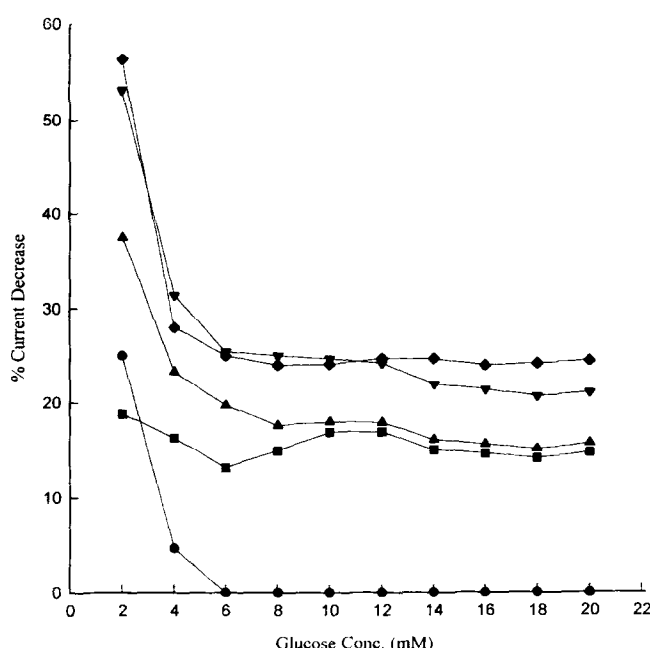


Figure 3 Results of exposure of three-layered electrodes to UV light. Percentage decrease in current output is plotted against glucose concentration for exposure times of: ●, 30 s; ■, 1 min; ▲, 2 min; ▼, 3 min; and ◆, 4 min.

Finally, the prepolymer-monomer mixture was applied to the electrodes and crosslinked in place. Each sensor was recalibrated after membrane formation and the results were compared to each electrode's original calibration curve. The calibration curves for one typical sensor are shown in Figure 5, and the results and statistical analysis for several electrodes are shown in Table 2. The average decrease in current response to glucose from 0 to 30 mM was $45 \pm 28\%$ at 37°C for films about 0.1 mm thick. The sensitivity changed from 1.27 ± 0.35 to $0.7 \pm 0.22 \text{ nA mM}^{-1}$.

ATR-FTIR was used to estimate the degree of crosslinking in the copolymer film by examining the degree of double bond conversion. The decrease in absorbance of the 1640 cm^{-1} peak, indicative of carbon-carbon double bond consumption, was approximately exponential in time after a brief linear period which lasted about 40 s, as shown in Figure 6. The degree of double bond conversion was estimated to be $95 \pm 4\%$ for a total UV exposure time of 2 min at

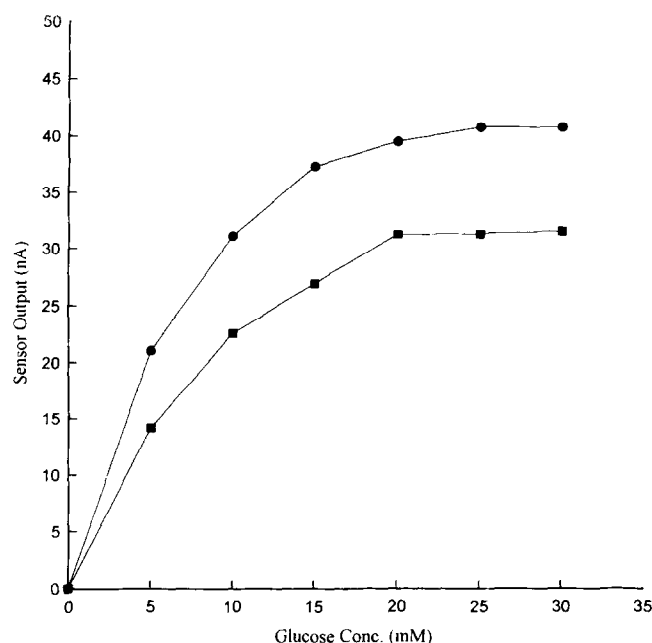


Figure 4 Results of exposure of three-layered electrodes to the monomer-prepolymer mixture. Output for one electrode is shown: ●, before and ■, after exposure to the solution.

200 mW cm^{-1} . The degree of swelling of the copolymer film in phosphate-buffered saline was evaluated by comparing dehydrated and hydrated mass measurements. The water content of the hydrated film samples was estimated to be $80 \pm 3\%$.

The subcutaneous response of the copolymer-treated sensors was evaluated in Sprague-Dawley rats. Figure 7 shows scanning electron micrographs of explanted sensors without (a) and with (b) treatment with the outer copolymer layer. The untreated electrode appeared to be completely encapsulated by fibrous tissue. The sensing portion of the electrode appeared to be completely isolated from the surrounding environment by the fibrous encapsulation. By contrast, the copolymer-treated sensor appeared to have very little fibrous tissue.

The interaction of a sensor with the tissue in contact with it is potentially detrimental to both the tissue and the sensor. Inflammation, swelling and necrosis of the tissue may be some of the results. It is important to maintain the sensitivity of the sensor. The sensor's

Table 1 Percentage reduction in sensor output at various glucose concentrations after exposure for 2 min to monomer-prepolymer-initiator solution

	Sensor 1	Sensor 2	Sensor 3	Sensor 4	Sensor 5	Mean	s.d.
Glucose conc. (mM):							
5	-58	-35	-126	14	33	-34	63
10	21	23	-4	12	27	16	12
15	17	17	16	2	28	16	9
20	21	21	5	2	21	14	10
25	21	22	5	2	23	15	10
30	24	20	4	1	23	14	11
Mean for all concentrations	8	11	-17	6	26		
Mean less 5 mM readings	21	21	5	4	24		
Overall mean (less 5 mM)	15						
Standard deviation	10						

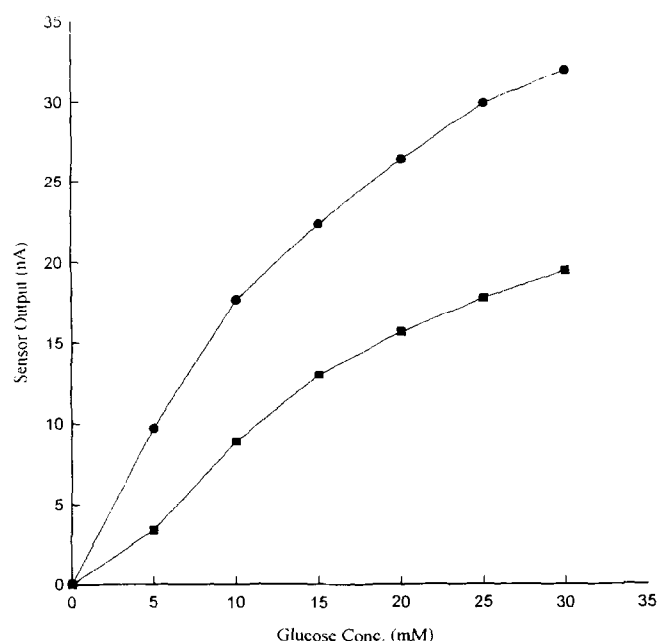


Figure 5 Results of application of HEMA-PEG-EDM copolymer membrane on three-layered electrodes. Output for one sensor: ●, before and ■, after application of the membrane.

output could be degraded by enzymes and reactive small molecules (e.g. H_2O_2 , superoxide) released by macrophages or other cells involved in foreign body response and through fouling by proteins contained in the subcutaneous fluid. The adsorption and attachment of proteins could lead to cell recognition and attachment and ultimately to encapsulation by fibrous material. The effect of enzymes, proteases and larger molecules could be limited by a molecular weight selective screen. Protein fouling could be mitigated by an outer material which exhibits relatively low protein adsorption.

There are several constraints that must be met in the selection of an outer biocompatible membrane for oxidoreductase-based sensors. The material, in either its final or applied form, must not cause excessive enzyme deactivation. This includes any steps needed to cure the membrane (if cured *in situ*). This is a particularly difficult constraint since many enzymes are susceptible to damage by exposure to organic solvents and heat and other energy sources (the

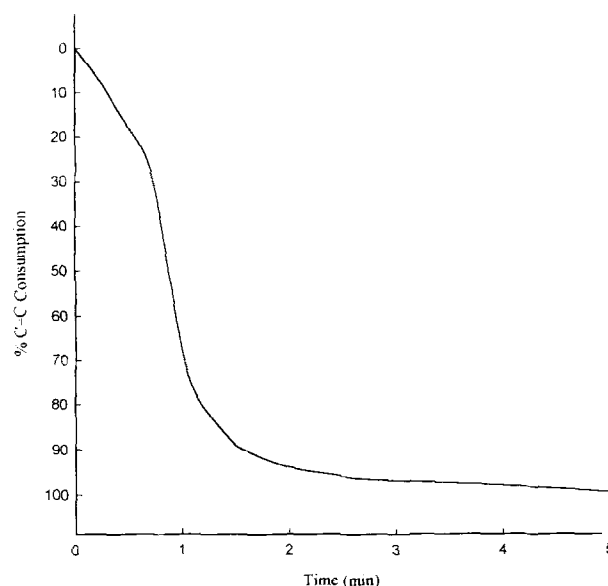


Figure 6 Approximate percentage of carbon-carbon double bond consumption in the copolymer as a function of UV light exposure duration. Percentage consumption values were obtained by comparing the 1640 cm^{-1} absorption values at $t = 0$, $t = 10\text{ min}$ and the value at any time t . These values were obtained from ATR-FTIR experiments.

monomer HEMA is itself an organic solvent and film crosslinking is initiated by a UV energy source). The material chosen must adhere well to that portion of the sensor system that it contacts. This could be an electrical insulation, a sensing layer film, or a film of some other active material. Finally, the membrane chosen must allow free transport of the analyte and the products of its transformation (in this system glucose and gluconolactone).

The HEMA-PEG-tetra-acrylate-ethylene dimethacrylate copolymer tested reduced encapsulation by fibrous tissue and did not cause any noticeable inflammation or tissue necrosis in rats. The copolymer described remained adherent to the material in layer 3 and the polyimide wire insulation for three days under the conditions of *in vivo* testing. The total loss in current was approximately 45%. Such loss is acceptable for a high current density sensor, such as the one described. The change in sensitivity of the membrane-coated sensors (1.27 to 0.70 nA mm^{-1}) is also considered acceptable for this high current density oxidore-

Table 2 Percentage reduction in sensor output at various glucose concentrations after application of HEMA-PEG-ethylene dimethacrylate copolymer membrane

	Sensor 1	Sensor 2	Sensor 3	Sensor 4	Mean	s.d.
Glucose conc. (mM):						
5	98	39	95	65	74	23
10	79	21	56	46	51	23
15	63	17	45	42	42	15
20	54	17	33	40	36	10
25	49	17	33	41	35	6
30	46	17	26	39	32	5
Averages	65	21	48	46	45	14
Overall average	45					
Standard deviation	18					

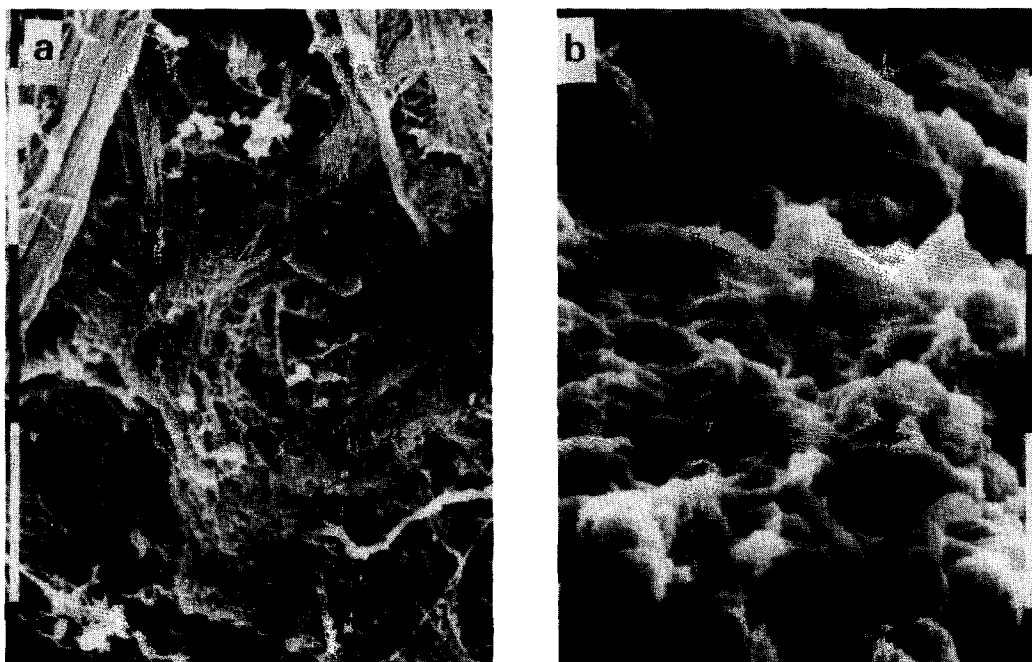


Figure 7 Scanning electron micrographs of copolymer untreated, **a**, and treated, **b**, electrodes implanted subcutaneously in Sprague-Dawley rats for 3 days. The untreated electrode appears to be highly encapsulated with fibrous tissue, while the copolymer-treated electrode appears to have much less fibrous encapsulation. The irregular texture of the copolymer-treated sensor is due to dehydration of the hydrogel membrane. Length bar represents 10 μm .

ductase-based sensor. The most important issue is the breadth of the linear range which is improved, although only slightly, by the addition of the copolymer membrane. Currents as low as 100 pA can reasonably be measured with the equipment used to drive these sensors. The losses were a function of UV light exposure to the light-sensitive enzyme, exposure to the monomer-prepolymer solution and mass transport limitations imposed by the hydrogel membrane.

Sensors of this design² afford a very high current density, typically 50–60 $\mu\text{A cm}^{-2}$ at 30 mM glucose in this study, prior to treatment. Accordingly, the 45% reduction in sensitivity caused by the application of the biocompatible membrane is entirely acceptable. Another feature of sensors of this design is their very fast response times, typically less than 60 s. Application of the biocompatible membrane would be expected to result in a minor decrease in responsiveness. This issue was not experimentally addressed in this study.

ACKNOWLEDGEMENTS

We acknowledge the financial support of the National Institutes of Health (DK-42015) and the National Science Foundation (BCS 9057641). We thank the Cell Research Institute at the University of Texas at Austin for the use of their scanning electron microscope and Mr Rob Lewis for his assistance with the microscopy. We thank Andrew Young for the use of his bipotentiostats. We thank Sten-Eric Lindquist and Elisabeth Csoregi for developing the etching technique, Ravi Rajagopalan and Tim O'Hara for synthesizing the redox polymer, Mark Vreeke for optimizing the interfer-

ence eliminating layer, and Elisabeth Csoregi and David Schmidtke for fabricating the sensors. We also thank Mark Vreeke and Mike Pishko for their assistance in conducting the ATR-FTIR experiments.

REFERENCES

- 1 Gregg BA, Heller A. Redox polymers containing enzymes. 1. A redox-conducting epoxy cement: synthesis, characterization, and electrocatalytic oxidation of hydroquinone. *J Phys Chem* 1991; **95**: 5970–5975.
- 2 Pishko MV, Michael AC, Heller A. Amperometric glucose microelectrodes prepared through immobilization of glucose oxidase in redox hydrogels. *Anal Chem* 1991; **63**: 2268–2272.
- 3 Bindra DS, Zhang Y, Wilson GS. Design and *in vitro* studies of a needle-type glucose sensor for subcutaneous monitoring. *Anal Chem* 1991; **63**: 1692–1696.
- 4 Poitout V, Moatti-Sirat D, Reach G, Zhang Y, Wilson G, Lemonnier F, Klein. A glucose monitoring system for on line estimation in man of blood glucose concentration using a miniaturized glucose sensor implanted in the subcutaneous tissue and a wearable control unit. *Diabetologia* 1993; **36**: 658–663.
- 5 Koudelka M, Rohner-Jeanrenaud F, Terrettaz J, Bobbioni-Harsch E, De Rooij N, Jeanrenaud B. *In vivo* behaviour of hypodermically implanted microfabricated glucose sensors. *Biosensors Bioelectron* 1991; **6**: 31–36.
- 6 Keedy FH, Vadgama P. Determination of urate in undiluted whole blood by enzyme electrode. *Biosensors Bioelectron* 1991; **6**: 491–499.
- 7 Mascini M, Selleri M, Riviello M. Glucose electrochemical probe with extended linearity for whole blood. *J Pharm Biomed Anal* 1989; **7**: 1507–1512.

- 8 Pickup JC, Shaw GS, Claremont DJ. *In vivo* molecular sensing in diabetes mellitus: an implantable glucose sensor with direct electron transfer. *Diabetologia* 1989; **32**: 213–217.
- 9 Gough DA, Armour JC, Lucisano JY, McKean BD. Short-term *in vivo* operation of a glucose sensor. *Trans Am Soc Artif Intern Organs* 1986; **XXXII**: 148–150.
- 10 Armour JC, Lucisano JY, McKean BD, Gough DA. Application of chronic intravascular blood glucose sensor in dogs. *Diabetes* 1990; **39**: 1519–1526.
- 11 Turner RF, Harrison DJ, Rajotte RV. Preliminary *in vivo* biocompatibility studies on perfluorosulphonic acid polymer membranes for biosensor applications. *Biomaterials* 1991; **12**: 361–368.
- 12 Brinkman E, Van Der Does L, Bantjes A. Poly(vinyl alcohol)-heparin hydrogels as sensor catheter membranes. *Biomaterials* 1991; **12**: 63–70.
- 13 Desai NP, Hubbell JA. Solution technique to incorporate polyethylene oxide and other water soluble polymers into surfaces of biomaterials. *Biomaterials* 1991; **12**: 144–153.
- 14 Sawhney AS, Hubbell JA. Poly(ethylene oxide)-graft-poly(L-lysine) copolymers to enhance the biocompatibility of poly(L-lysine) alginate microcapsule membranes. *Biomaterials* 1992; **13**: 863–870.
- 15 Grainger DW, Nojiri C, Okano T, Kim SW. *In vitro* and *ex vivo* platelet interactions with hydrophilic-hydrophobic poly(ethylene oxide)-polystyrene multiblock copolymers. *J Biomed Mater Res* 1989; **23**: 979–1005.
- 16 Gombotz WR, Guanghui W, Hoffman AS. Immobilization of poly(ethylene oxide) on poly(ethylene terephthalate) using a plasma polymerization process. *J Appl Polym Sci* 1989; **37**: 91.
- 17 Pathak CP, Sawhney AS, Hubbell JA. Rapid photopolymerization of immunoprotective gels in contact with cells and tissue. *J Am Chem Soc* 1992; **114**: 8311–8312.
- 18 Csoregi E, Quinn CP, Schmidtke DW *et al.* Design, characterization and one-point *in vivo* calibration of a subcutaneously implanted glucose electrode. *Anal Chem* 1994; **66**: 3131–3138.
- 19 Maidan R, Heller A. Elimination of electrooxidisable interferant-produced currents in amperometric biosensors. *Anal Chem* 1992; **64**: 2889–2896.